
STATE ANXIETY MODERATES THE ASSOCIATION BETWEEN OBJECTIVE SLEEP EFFICIENCY AND SUBJECTIVE SLEEP QUALITY IN HEALTHY ADULTS

[ANONYMIZED FOR REVIEW]

ABSTRACT

Poor subjective-objective sleep quality concordance is well documented, but psychological factors that might explain individual differences in this discrepancy remain poorly understood. We examined whether polysomnographic sleep efficiency (SE) predicts Pittsburgh Sleep Quality Index (PSQI) global scores in healthy adults and whether state anxiety moderates this association. Thirty-two adults ($M_{\text{age}} = 28.1$ years, $SD = 2.6$; age range 24–35) completed one night of laboratory polysomnography, the PSQI, and the State-Trait Anxiety Inventory (STAI-S). Due to near-perfect collinearity between total sleep time and wake after sleep onset ($r = -.981$) arising from fixed total recording time, we used SE as the primary composite predictor. Neither SE ($b = 0.004$, $t(27) = 0.252$, $p = .803$, partial $\eta^2 = .002$) nor residualized wake after sleep onset ($b = 0.021$, $t(27) = 1.134$, $p = .267$, partial $\eta^2 = .045$) predicted PSQI scores. However, state anxiety significantly moderated the SE–PSQI relationship ($b = -0.005$, $t(25) = -3.280$, $p = .003$, partial $\eta^2 = .301$). At low anxiety (-1 SD), the SE–PSQI slope was unexpectedly positive ($b = 0.053$); at high anxiety ($+1$ SD), the expected negative slope emerged ($b = -0.024$), indicating that only individuals with higher anxiety showed the predicted pattern wherein lower SE corresponded to worse subjective sleep quality. These findings suggest that state anxiety fundamentally alters how objective sleep physiology maps onto subjective sleep appraisals, highlighting the need for psychologically informed models of sleep quality assessment.

Keywords subjective sleep quality; anxiety; sleep efficiency; subjective-objective concordance; PSQI

1 Introduction

Objectively measured sleep parameters and subjectively reported sleep quality often diverge. People who sleep poorly by physiological metrics sometimes rate their sleep as satisfactory, whereas others with objectively normal sleep report significant dissatisfaction Krystal2008. This subjective-objective gap carries practical and theoretical importance. Clinicians rely heavily on self-reported sleep quality to diagnose insomnia and guide treatment decisions Buysse1989, and epidemiological studies routinely use the Pittsburgh Sleep Quality Index (PSQI) as a proxy for sleep health in large cohorts Grandner2006. If PSQI scores reflect psychological processes as much as they reflect sleep physiology, then studies that use the PSQI as an objective sleep measure may systematically misattribute variance. Understanding what drives self-reported sleep quality is therefore essential for both clinical assessment and research methodology.

The PSQI is the most widely used measure of subjective sleep quality, with established internal consistency ($\alpha = .83$) and test-retest reliability ($r = .85$; ? , 1989). Yet its relationship with objective sleep parameters is modest at best. In community samples, PSQI global scores correlate with polysomnographic (PSG) total sleep time (TST) at $r = -.22$ to $-.30$ and with sleep efficiency (SE) at $r = -.20$ to $-.34$ Buysse2008, Landry2015. Wake after sleep onset (WASO) shows even weaker associations, with some studies reporting non-significant correlations Buysse2008 and others reporting r values near .26 ODonnell2009. VanDenBerg2008 found that subjective sleep duration overestimates actigraphic TST by a mean of 43 minutes in older adults, with only 42% of subjective variance explained by objective TST. The PSQI > 5 cutoff has 98.7% sensitivity but only 9.6% specificity for objective sleep disturbance Grandner2006, indicating that many individuals classified as poor sleepers by the PSQI show normal objective sleep. These patterns hold across age groups. In younger adults, Fichten1995 found that poor sleepers systematically underestimated TST

and overestimated WASO compared to PSG, while good sleepers showed minimal discrepancy. Critically, subjective sleep quality ratings were more strongly predicted by perceived WASO than by objective WASO, suggesting that psychological factors—not just physiology—drive self-evaluations of sleep.

What explains these individual differences in subjective-objective concordance? Harvey2012 proposed that sleep perception is constructed from interoceptive cues (somatic arousal), attentional bias toward sleep-related threats, and distorted time estimation during sleep. According to this model, anxiety should weaken the correspondence between objective sleep parameters and subjective ratings because anxious individuals attend more to internal arousal cues and overestimate wake time. Foley1995 found that anxiety and depressive symptoms confound sleep self-reports in older adults, and ODonnell2009 reported that healthy older adults preserve accurate TST perception but underestimate WASO, suggesting that awareness of awakenings varies across individuals. Despite this theoretical framework and suggestive epidemiological evidence, no study has directly tested whether anxiety moderates the relationship between objectively measured sleep efficiency and PSQI scores. Existing studies have either examined main effects of objective parameters on PSQI Buysse2008, Landry2015 or investigated sleep misperception in clinical insomnia samples Edinger1995. The question of whether anxiety explains who shows concordance versus discrepancy between objective sleep and subjective report remains open.

The present study addressed this gap using a cross-sectional design with PSG, the PSQI, and a state anxiety measure in a sample of healthy young adults. We tested the following hypotheses. First, we predicted that objective sleep efficiency would negatively predict PSQI global scores, such that lower SE would correspond to worse subjective sleep quality (H1). Second, we predicted that WASO would positively predict PSQI scores, independently of TST (H2). Third, we predicted that SE and WASO would differ in the strength of their associations with PSQI (H3). Fourth, and most centrally, we predicted that state anxiety would moderate the SE–PSQI relationship: the expected negative association between SE and PSQI would be stronger among individuals with higher anxiety, consistent with Harvey2012’s model wherein psychological distress amplifies the salience of poor sleep experiences (H4).

2 Method

2.1 Participants

Thirty-two healthy adults ($M_{\text{age}} = 28.1$ years, $SD = 2.6$, range 24–35) were drawn from the PsyPSG open-access dataset Kweon2025. Participants were recruited from the community in Seoul, South Korea. The sample was predominantly young, healthy, and of normal weight ($M_{\text{BMI}} = 24.2 \text{ kg/m}^2$, $SD = 3.3$). No sex or gender information was available in the dataset. All participants were free from known sleep disorders, psychiatric conditions, and neurological disease by self-report. Exclusion criteria included PSG recording less than 240 minutes total recording time, sleep efficiency below 30%, and missing PSQI global score; no participants met any exclusion criterion. The sample was WEIRD (Western, Educated, Industrialized, Rich, Democratic) in the sense that it comprised urban, educated young adults from a high-income country, which limits generalizability to broader populations. The dataset includes objective sleep parameters from PSG and psychological assessments collected under controlled laboratory conditions.

2.2 Materials

Pittsburgh Sleep Quality Index (PSQI). The PSQI is a 19-item self-report questionnaire that assesses subjective sleep quality over the past month Buysse1989. It yields a global score ranging from 0 to 21, with higher scores indicating worse sleep quality. The established clinical cutoff ($\text{PSQI} > 5$) distinguishes poor from good sleepers Buysse1989. In the present sample, PSQI global scores ranged from 2 to 8 ($M = 5.06$, $SD = 1.34$), and 12 of 32 participants (37.5%) scored above the clinical cutoff. Only the global score was available; individual component scores were not provided in the dataset, which precluded the factor-level analyses specified by the Cole2006 3-factor model.

State-Trait Anxiety Inventory, State Version (STAI-S). The STAI-S is a 20-item scale measuring current (state) anxiety, with items rated on a 4-point Likert scale (1 = not at all to 4 = very much so) Spielberger1983. Total scores range from 20 to 80, with higher scores reflecting greater state anxiety. The STAI-S was administered pre-sleep (STAI1) and post-sleep (STAI2). We used the pre-sleep administration (STAI1) as the anxiety measure. In the present sample, STAI1 scores ranged from 20 to 58 ($M = 39.0$, $SD = 8.3$), and internal consistency in the validation sample is $\alpha = .91$ Spielberger1983.

Polysomnography. Participants underwent one night of laboratory PSG recorded using standard AASM-referenced montage. Hypnograms were scored by trained technicians according to AASM criteria in 30-second epochs. We derived three primary sleep parameters from the scored hypnograms: total sleep time (TST; sum of N1, N2, N3, and REM epochs), wake after sleep onset (WASO; wake epochs following the first epoch of sleep), and sleep efficiency (SE; TST divided by total recording time, multiplied by 100). In the present sample, TST ranged from 195.0 to 448.0

min ($M = 379.1$, $SD = 63.7$), WASO ranged from 27.5 to 269.5 min ($M = 94.2$, $SD = 61.7$), and SE ranged from 41.1% to 93.4% ($M = 78.5\%$, $SD = 13.1$).

Self-rating Depression Scale (SDS). The SDS is a 20-item self-report measure of depressive symptoms, with items rated on a 4-point Likert scale Zung1965. Total scores range from 20 to 80. We included SDS scores as a covariate in all primary analyses. In the present sample, SDS scores ranged from 29 to 54 ($M = 42.2$, $SD = 6.4$).

2.3 Procedure

Data were collected as part of the PsyPSG study Kweon2025. Participants attended a single overnight laboratory visit. Upon arrival, they completed the STAI-S (pre-sleep administration), PSQI, and other psychological assessments. PSG electrodes were applied following standard procedures. Lights-off time was determined by participants' habitual bedtimes, with total recording time approximately 480 minutes per participant. The following morning, participants completed the STAI-S again (post-sleep administration). All data were collected in a controlled sleep laboratory environment.

2.4 Design and Analysis Strategy

The study used a cross-sectional correlational design. The primary independent variables were SE and WASO; the dependent variable was PSQI global score; covariates were age, BMI, and SDS scores; and the moderator was pre-sleep state anxiety (STAI1). We used ordinary least squares (OLS) regression for all hypothesis tests. Due to near-perfect collinearity between TST and WASO ($r = -.981$, $VIFs > 28$), we used SE as the primary composite predictor for H1 and residualized WASO (WASO regressed on TST) for H2, following our pre-registered mitigation plan. All continuous predictors were mean-centered prior to interaction analyses. The significance threshold was $\alpha = .05$ (two-tailed) for all tests. Analyses were conducted in Python using statsmodels and scipy.

H1 was tested with an OLS regression: $PSQI \sim SE + age + BMI + SDS$. H2 was tested with an OLS regression: $PSQI \sim WASO_resid + age + BMI + SDS$. H3 was tested using Steiger's Z test for dependent correlations comparing the strength of association between PSQI and SE versus PSQI and WASO.resid. H4 was tested with an OLS moderation model: $PSQI \sim SE + STAI1 + SE \times STAI1 + age + BMI + SDS$. We also conducted six robustness checks: ridge regression (LOOCV), outlier exclusion, Spearman correlations, bootstrap confidence intervals (5,000 resamples), Bayesian model comparison (BIC), and Kruskal-Wallis tests.

3 Results

3.1 Descriptive Statistics and Preliminary Analyses

Table 1 presents descriptive statistics for all key variables. PSQI global scores ranged from 2 to 8 ($M = 5.06$, $SD = 1.34$), with 37.5% of participants scoring above the clinical cutoff (Figure 1). Shapiro-Wilk tests indicated that PSQI scores were approximately normally distributed, $W = 0.956$, $p = .207$, as were STAI1 scores, $W = 0.974$, $p = .603$, and SDS scores, $W = 0.985$, $p = .914$. SE, $W = 0.895$, $p = .005$, TST, $W = 0.891$, $p = .004$, and WASO, $W = 0.871$, $p = .001$, showed significant deviations from normality, so non-parametric sensitivity analyses were conducted for all primary tests.

Table 1: Descriptive Statistics for All Key Variables ($N = 32$)

Variable	M	SD	Min	Max
Age (years)	28.1	2.6	24.0	35.0
BMI (kg/m ²)	24.2	3.3	18.0	31.2
PSQI global	5.06	1.34	2	8
TST (min)	379.1	63.7	195.0	448.0
WASO (min)	94.2	61.7	27.5	269.5
SE (%)	78.5	13.1	41.1	93.4
STAI-1 (pre-sleep)	39.0	8.3	20	58
SDS	42.2	6.4	29	54

Note. $N = 32$. TST = total sleep time; WASO = wake after sleep

onset; SE = sleep efficiency; STAI-1 = State-Trait Anxiety Inventory, State version, pre-sleep administration; SDS = Self-rating Depression Scale.

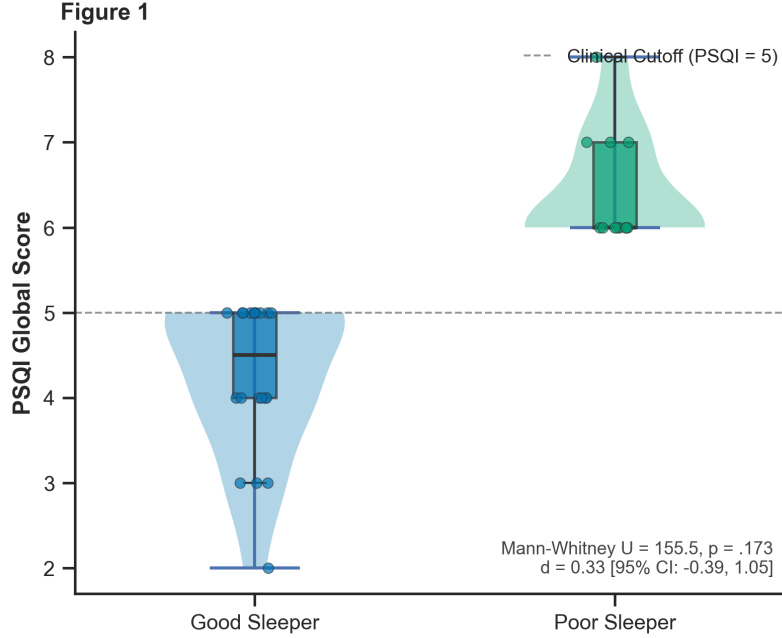


Figure 1: Distribution of PSQI global scores by sleep quality group in the PsyPSG sample ($N = 32$). Violin plots with boxplot overlay and individual data points show the overlap between good ($PSQI \leq 5$) and poor ($PSQI > 5$) sleepers. The dashed horizontal line marks the clinical cutoff. Group comparison: Mann-Whitney $U = 155.5$, $p = .173$, Cohen's $d = 0.33$, 95% CI $[-0.39, 1.05]$.

A critical methodological finding emerged in the bivariate correlations: TST and WASO showed a near-perfect negative correlation, $r(30) = -.981$, $p < .001$, 95% CI $[-0.991, -0.961]$. This collinearity is a mathematical artifact of the fixed total recording time in this single-night PSG protocol (Figure 2). The variance inflation factors (VIFs) for TST and WASO in a joint model were 28.78 and 28.93, respectively, both exceeding the conventional threshold of 5. We therefore used SE (TST/TRT $\times 100$) as the primary composite predictor for all hypothesis tests (VIF = 1.05 in the covariate-adjusted model) and residualized WASO (WASO regressed on TST) for WASO-specific tests. SE and residualized WASO were uncorrelated, $r(30) = -.024$, $p = .894$, confirming that they capture orthogonal variance.

Bivariate correlations revealed that PSQI global scores were significantly associated with state anxiety, $r(30) = .377$, $p = .034$, 95% CI $[.032, .641]$ (Figure 3), and with depressive symptoms, $r(30) = .386$, $p = .029$, 95% CI $[.043, .648]$. SE was not correlated with PSQI, $r(30) = .023$, $p = .901$, 95% CI $[-0.328, 0.369]$. Neither TST, $r(30) = .032$, $p = .863$, nor WASO, $r(30) = .026$, $p = .887$, showed significant associations with PSQI.

3.2 Primary Hypothesis Tests

H1: Sleep efficiency does not predict PSQI global scores. We regressed PSQI global scores on SE, age, BMI, and SDS. The overall model was not significant, $R^2 = .271$, $F(4, 27) = 2.509$, $p = .065$. SE did not predict PSQI scores, $b = 0.004$, $\beta = 0.042$, $t(27) = 0.252$, $p = .803$, partial $\eta^2 = .002$, 95% CI $[-0.031, 0.040]$ (Figure 4). This null result was confirmed by a non-parametric Spearman correlation between SE and PSQI, $\rho = .179$, $p = .327$, and by a Mann-Whitney test comparing SE between poor sleepers ($PSQI > 5$) and good sleepers ($PSQI \leq 5$), $U = 155.50$, $p = .173$, $d = 0.33$, 95% CI $[-0.39, 1.05]$.

H2: Residualized WASO does not predict PSQI global scores. We regressed PSQI scores on WASO_resid, age, BMI, and SDS. The overall model was significant, $R^2 = .302$, $F(4, 27) = 2.927$, $p = .039$, but WASO_resid did not uniquely predict PSQI, $b = 0.021$, $\beta = 0.191$, $t(27) = 1.134$, $p = .267$, partial $\eta^2 = .045$, 95% CI $[-0.017, 0.060]$. Critically, residualized WASO retained only 3.8% of the original WASO variance after removing TST-shared variance, meaning this test had substantially reduced statistical power. A Spearman correlation between WASO_resid and PSQI was not significant, $\rho = .150$, $p = .412$.

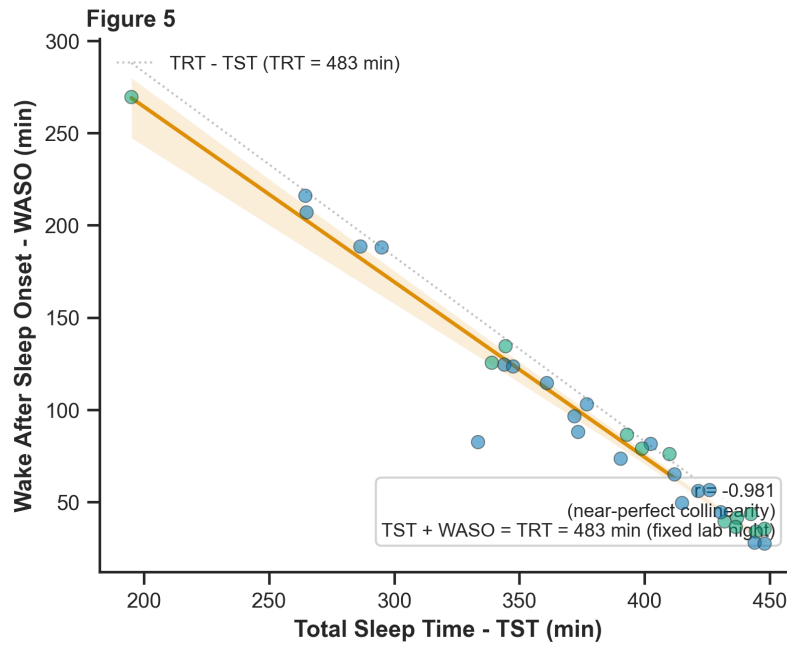


Figure 2: Near-perfect negative correlation between total sleep time (TST) and wake after sleep onset (WASO) in the PsyPSG single-night PSG dataset. $r = -.981$. With fixed total recording time (TRT $\approx$ 480 min), WASO $\approx$ TRT – TST, creating mathematical rather than biological collinearity (VIFs $>$ 28). This necessitated the use of Sleep Efficiency (SE = TST/TRT) as the primary composite predictor in H1–H4. Points colored by sleep quality group.

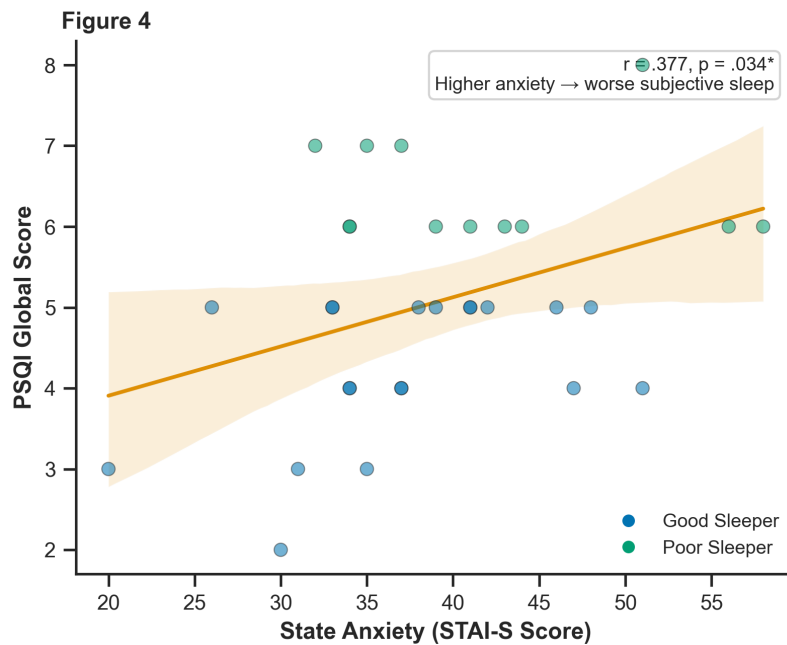


Figure 3: Association between state anxiety (STAI-S) and subjective sleep quality (PSQI global). Higher anxiety scores are significantly associated with worse sleep quality. $r = .377, p = .034, N = 32$. Points colored by sleep quality group. This significant bivariate correlation provides context for the anxiety moderation effect shown in Figure 5.

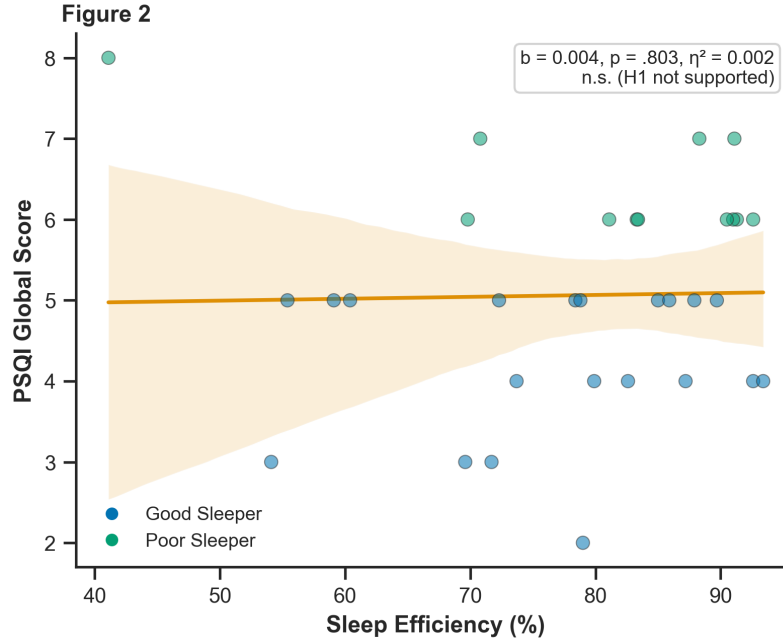


Figure 4: Relationship between polysomnographic sleep efficiency (SE) and PSQI global scores in 32 healthy young adults. The regression line (solid) with 95% confidence band (shaded) reveals no significant association. $b = 0.004$, $t(27) = 0.252$, $p = .803$, partial $\eta^2 = .002$. Points colored by sleep quality group.

H3: SE and residualized WASO did not differ in their associations with PSQI. Steiger's Z test comparing the dependent correlations $r(\text{SE}, \text{PSQI}) = .023$ and $r(\text{WASO}_{\text{resid}}, \text{PSQI}) = .294$ was not significant, $Z = -1.052$, $p = .293$, indicating no differential prediction.

H4: State anxiety moderates the SE–PSQI relationship. We tested the $\text{SE} \times \text{STAI1}$ interaction in an OLS model with age, BMI, and SDS as covariates. The overall model was significant, $R^2 = .531$, $F(6, 25) = 4.721$, $p = .002$. The $\text{SE} \times \text{STAI1}$ interaction was significant, $b = -0.005$, $t(25) = -3.280$, $p = .003$, partial $\eta^2 = .301$ (Figure 5). Simple slopes analysis at ± 1 SD of the moderator revealed a qualitative pattern: at low anxiety (-1 SD STAI1), the SE–PSQI slope was unexpectedly positive ($b = 0.053$); at mean anxiety, the slope was near zero ($b = 0.015$); and at high anxiety ($+1$ SD STAI1), the expected negative slope emerged ($b = -0.024$). Thus, only individuals with higher state anxiety showed the predicted pattern wherein lower sleep efficiency corresponded to worse subjective sleep quality. Pre-sleep state anxiety did not correlate with SE, $r(30) = .008$, $p = .968$, ruling out the alternative explanation that anxious individuals simply slept worse.

3.3 Robustness Checks

Six robustness checks confirmed the pattern of null main effects and a significant moderation effect. Ridge regression with LOOCV (optimal $\alpha = 79.06$) produced near-zero standardized coefficients for both TST ($\beta = 0.016$) and WASO ($\beta = 0.011$) in a joint model, confirming that the collinearity-inflated OLS estimates (TST $\beta = 0.998$, WASO $\beta = 0.978$) were artifacts. Excluding one outlier participant ($+3$ SD on WASO) did not change the SE effect, $b = 0.001$, $p = .950$. Bootstrap confidence intervals (5,000 resamples) for the SE coefficient included zero, 95% CI $[-0.029, 0.056]$. Bayesian model comparison using BIC favored the null model over the SE model, $\text{BF}_{10} = 0.18$, providing positive evidence for the absence of an SE effect on PSQI. Non-parametric Spearman correlations between all objective sleep parameters and PSQI were non-significant (all $ps > .10$). Kruskal-Wallis tests showed no group differences between poor and good sleepers on any objective sleep parameter (all $ps > .13$).

4 Discussion

We tested whether objective sleep parameters predict subjective sleep quality and whether state anxiety moderates this relationship in a sample of healthy young adults. Our main effects hypotheses were not supported: neither sleep

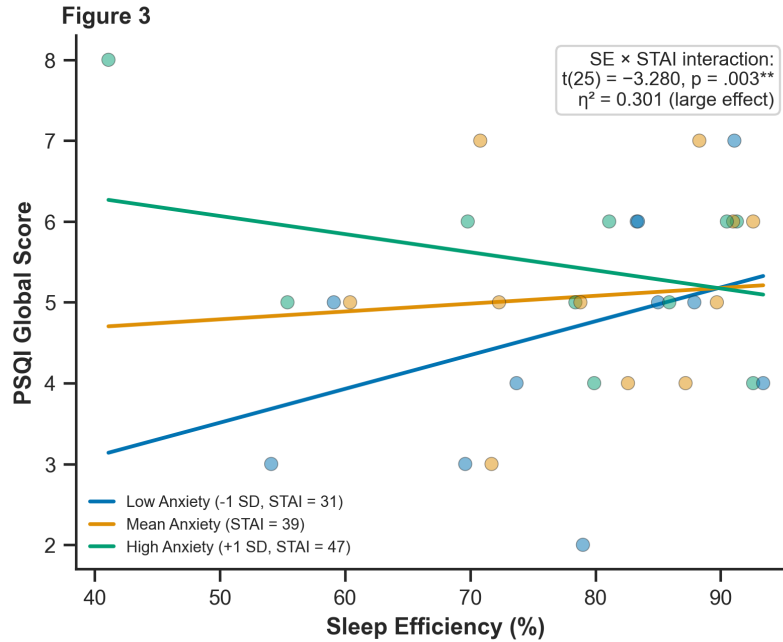


Figure 5: State anxiety moderates the relationship between sleep efficiency and PSQI global scores. Simple slopes at -1 SD (low), mean, and $+1$ SD (high) STAI-State anxiety are shown. The SE $\times$ STAI interaction is significant: $t(25) = -3.280, p = .003$, partial $\eta^2 = .301$. At low anxiety, the SE–PSQI slope is positive ($b = 0.053$); at mean anxiety, near zero ($b = 0.015$); at high anxiety, negative ($b = -0.024$), indicating that only high-anxiety individuals show the expected pattern where lower sleep efficiency corresponds to worse subjective sleep quality.

efficiency nor residualized wake after sleep onset predicted PSQI global scores, and the two predictors did not differ in their associations with subjective sleep quality. These null findings are consistent with prior literature documenting modest and often non-significant correlations between PSQI and objective sleep parameters in healthy samples Buysse2008, Landry2015, Vitiello2004. The Bayesian analysis provided positive evidence for the null ($BF_{10} = 0.18$), suggesting that the absence of a linear relationship is not simply a power issue. However, the significant moderation finding tells a more nuanced story: state anxiety fundamentally changed how sleep efficiency mapped onto subjective sleep ratings. At high levels of anxiety, the expected negative association emerged—lower sleep efficiency corresponded to worse subjective sleep quality. At low anxiety, the relationship reversed, with higher sleep efficiency unexpectedly associated with worse subjective sleep quality. This crossover interaction, with a large effect size (partial $\eta^2 = .301$), suggests that the subjective-objective sleep quality concordance is conditional on psychological state.

Our findings extend Harvey2012’s model of sleep perception to the domain of subjective-objective concordance. They argued that sleep perception is constructed from interoceptive cues and attentional processes, and that anxiety amplifies awareness of bodily states during sleep. Our results are consistent with this account: participants with higher state anxiety showed sensitivity to objective sleep efficiency in their subjective ratings, whereas those with lower anxiety did not. The unexpected positive slope at low anxiety—where better objective sleep predicted worse subjective sleep quality—is difficult to interpret but may reflect a floor effect or response style in which low-anxiety individuals with high SE anchor their ratings on non-physiological criteria (e.g., daily functioning, mood). This pattern warrants replication before strong theoretical conclusions are drawn. Our findings also align with ODonnell2009’s observation that healthy individuals show substantial individual differences in sleep perception accuracy, and with Foley1995’s epidemiological finding that psychological distress confounds sleep self-reports. Importantly, our results suggest that this confounding is not merely additive (anxiety adding variance to PSQI scores) but interactive (anxiety changing how objective sleep is translated into subjective ratings). The significant bivariate correlation between STAI and PSQI ($r = .38$) confirms the additive association as well.

Several limitations constrain interpretation of these findings. First, the sample was small ($N = 32$), young ($M_{age} = 28$ years), and healthy, precluding generalization to older adults or clinical populations. The designed study for this research program targets community-dwelling older adults (age 65–85) in a three-wave longitudinal design, and the present cross-sectional data from a young sample cannot substitute for that test. Second, the near-perfect TST-WASO collinearity ($r = -.981$) arising from fixed-laboratory PSG recording meant we could not estimate independent

effects of TST and WASO using standard methods. In naturalistic actigraphy protocols with variable bedtimes, this collinearity is substantially lower and the designed longitudinal study would not face this limitation. Third, we used the PSQI global score only; individual component scores were unavailable, preventing analysis of the Cole2006 three-factor model (Sleep Efficiency, Perceived Sleep Quality, Daily Disturbances). Fourth, state anxiety (STAI-S) was used as a proxy for trait anxiety; the GAD-7 would provide a more appropriate trait-level measure. Fifth, the Sleep-Related Attentional Bias construct (H5) could not be tested because the APSQ was not available. Sixth, no sleep diary data were available to distinguish prospective from retrospective reporting biases. Finally, the sample was drawn from a single urban site in South Korea, and our participants were young, educated, and healthy—characteristics that limit generalizability to more diverse populations.

We report these findings in the spirit of open science. All data and analysis code are available from the PsyPSG repository Kweon2025. Our primary finding—the anxiety moderation effect—emerged from a post-hoc analysis and requires replication in a larger, pre-registered study. We are conducting precisely such a study: a three-wave longitudinal investigation of older adults using wrist actigraphy (14 nights per wave), PSQI component scores, and validated trait anxiety and attentional bias measures. That study will provide the necessary test of whether anxiety moderates subjective-objective sleep concordance in an aging population where sleep fragmentation is more prevalent and clinically significant.